JusticeLens: A Hybrid ML–XAI Approach for Indian Bail Judgment Prediction Using TF-IDF and Structured Metadata

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***Abstract*— India's judicial process of granting bail has evolved over time into an elaborate negotiation of societal and legal reasoning influenced by the applicant's gender, criminal antecedents, and the very nature of the offense. In light of that, we hereby introduce an integrated machine-learning framework-conceived to predict bail-positive or bail-negative outcomes, given some structured legal attributes and unstructured summaries of judgments-called JusticeLens. The dataset residing in MongoDB is subjected first to preprocessing involving normalization, encoding of categorical variables, and rectification of any class imbalance introduced through SMOTE. The structured features of gender, prior cases, and crime type are treated as features alongside judgment text representations TF-IDF processed. The prediction model is based on the optimized XGBoost classifier hyper-parameter tuned with Grid Search Cross Validation, later ensemble with Logistic Regression and Random Forest to enhance its predictive performance further, validated using weighted F1 metrics. To satisfy the requirement of interpretability, SHAP is used to highlight the key legal variables. The framework illustrates how explainable AI can contribute to a transparent, accountable, and data-driven analysis of bail decisions within the Indian judiciary.**

***Keywords—Judicial analytics, Bail prediction, Machine learning, XGBoost, Explainable AI, SHAP, Natural language processing, TF–IDF, Ensemble learning, MongoDB, Legal informatics*.**

I. Introduction

Granting bail or refusing it is one of the foundational decision points for liberty in an Indian judicial system, while at the same time upholding the tenets of justice against individuals. Such bail determinations depend a great deal on a mixture of legal and factual considerations and to some extent subjective elements, such as severity of the offence, past criminal record, social background, and even gender. Therefore, the assessment of merit in bail applications sometimes turns out to be inconsistent and prone to bias, which increases the need for the exploration of data-based solutions to make the bail decision process fairer, more transparent, and efficient. [4, 13].

Of late, artificial intelligence (AI) and machine learning (ML) have shown very much promising development in such new technologies in the field of legal analytics. That is those technologies that automatically disclose hidden patterns and associations among attributes of the historical database and predictable insights toward humans. Subject to the definition of computational law, a combination of data science with law studies, personalized predictive modeling in domains like the prediction of case outcome forecasting, or judgment summarization. Bail prediction, although unexplored with respect to explainable AI techniques, is especially true within the Indian scenario and its intrinsic challenges of unstructured judgments and heterogeneous data sources. JusticeLens-a hybrid predictive analytics framework that processes Indian bail judgments to predict the likelihood of awarding or denying bail-is what this study attempts to answer. The system is built by combining structured demographic and legal parameters such as gender, crime type, and other prior cases with unstructured judicial text from case summaries. While attributes generated through structured methods capture measurable attributes, they ignore qualitative inputs from textual data, which would account for vocabulary that signify judicial reasoning[3, 12, 14]. The proposed process integrates both forms of data in mitigating the limitations of models that depend solely on numerical or textual input.

Collection and storage of the data begin using MongoDB, followed by full pre-processing on textual values and categorical values to normalize them. Feature engineering enhancement is that interaction variables such as risk score and gender-crime interaction are introduced to capture richer relational patterns through the data. Judgment text has been transformed to numerical vectors using the Term Frequency–Inverse Document Frequency (TF–IDF) method, helping the model understand linguistic features based on statistical counts [1, 2, 8] An Extreme Gradient Boosting (XGBoost) is then trained and optimized by Grid Search Cross-Validation for the ideal hyperparameters. Finally, predictions will be interpreted through the use of SHapley Additive explanations (SHAP) to keep the model explainable and transparent in the legal sense. The system has shown that the integration of explainable AI with legal datasets really isn't about improving their predictive accuracy, but can also lead to better decision making along increasingly ethical and evidence-based lines within the judiciary. This framework can be used by legal practitioners as a decision-support system, indicating how factors such as gender, presence of prior records, and crime types affect bail decisions made in Indian courts [1, 10]

A. Problem Statement

Even though a plethora of Indian court judgments is available, the majority of the data is in unstructured format, thus making the same arduous in extracting meaning patterns by traditional means. Presently available legal analytics tools

depend chiefly on keyword searches or rule-based systems and neglect the deeper links between textual reasoning and structured case attributes, such as gender, severity of offense, and prior criminal history. Moreover, many predictive models presently in use are black boxes, low on interpretability and accountability. The objective is to develop an explainable machine learning framework for predicting bail outcomes integrated with structured legal features and unstructured judgment text, achieving both accuracy and interpretability for judicial decision support systems. The focus of this system is to narrow the retrogression that lies between predictive mechanism and explainable AI in an effort to enable stakeholders to have a transparent data-backed illumination of the patterns of bail judgments.

II. Literature Survey

Legal analytics and AI have become increasingly evolved in the direction of enhanced transparency, consistency, and efficiency in judiciary systems. Early studies were primarily based on classical machine-learning algorithms with structured and textual legal datasets. For instance, the study in SAGE Open showed how a case-theoretics study could predict outcomes of the Indian Supreme Court when combined with TF-IDF and models such as Decision Trees, Random Forests, and SVMs, as preprocessing of text and normalization to domain-specific terms were some muscle knots in predictive performance. The early approaches prepared ground for more viable computational legal analysis in that country. And so, transformer-based methods have taken the storm. The Legal Knowledge Extraction and Prompt Learning (LKEPL) framework published in the *Journal of King Saud University – Computer and Information Sciences* fused factual descriptions with statutory information using prompt-based transformer embeddings on the CAIL2018 dataset. LKEPL turned legal semantics into self-attention and knowledge prompts and achieved generalization in low-resource situations, shifting the level of analysis from surface text features to deep semantic representations.

The growing concern for explainability in legal AI brought exhaustive reviews such as [3] in *Human-Centric Intelligent Systems*, which discussed intrinsic and post-hoc XAI techniques, highlighting the need for balance between a prediction power and interpretability in law. The complementary study of [4] described XAI-based legal models that weigh feature importance, rationale extraction, and transparency mechanisms key to trust by judges. All these works indicated that explainability is central-not optional-in public-facing AI systems. Along such improvements, individual predictions of judgments by courts have been strongly bolstered using ensemble learning. A study published in *Applied Sciences* [5] used Random Forests and Gradient Boosting with domain-engineered features like statutory references and case metadata to Indian high-court decisions, outperforming text-only models and proving effectiveness of hybrid structured-text approaches. As for [6], heterogeneous graph representations emerged capturing relationships among facts, statutes, and charges. Graph Neural Networks were used to improve performance on CAIL2018 through a relational structure embedding beyond a linear text [7]. The proposed Hierarchical Graph Networks (HGN) to integrate document-level hierarchy-facts, reasoning, judgment-with statutory-article relationships, achieving gains in accuracy and interpretability. In the United States, [8] constructed benchmark class-action litigations that relied on mixing transformer and graph model representations while further validating the utility of relational reasoning in complex adjudication tasks. These, taken together, witness the evolution from shallowly feature-observing systems to deep structures aware architectures that equate with human legal reasoning.

Literally, explainability and accountability have become the cornerstones of judicial AI. Meta-review in [9] synthesized over 200 studies over law, health, and governance and submitted the XAI as a core requirement in high-stakes domains. Besides this, [10] in *AI & Society* takes it forward by developing a typology of algorithmic autonomy in legal reasoning and documents the movement of rule-based models to statistical and deep learning, once again igniting all arguments around oversight, ethics, and human interpretative control. Many more recent actions revolve around linguistic and regional diversities. The HLDC [11] includes 90,000 Hindi bail judgments (2019-2021), which produce a 93% accurate bail prediction using CNNs and an ad under the line of non-English judicial analytics. In the same vein, the ILDC [12] captured labeled Indian judgments along with outcome and rationale sentences for both prediction and explanation use-cases. Socio-legal analyses like [13] examined how actuarial justice was transitioning and amplified concerns on bias by automated pre-trial risk assessments by advocating for ethical frameworks around judicial AI. Some limitations of [14] include overfitting, small-sample bias, and inadequate domain interpretability, which enhances the need for rigorous legal reasoning to go together with data-driven modeling.

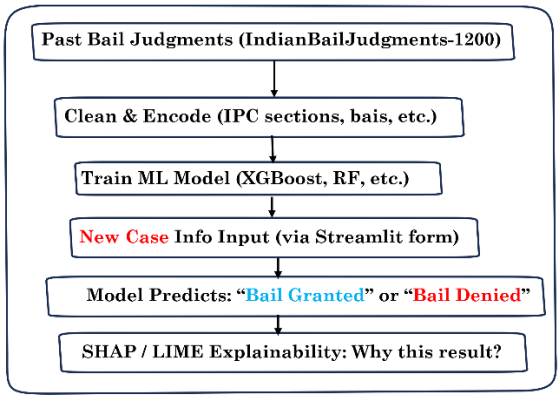


Fig.1 Logic Flow Diagram

To fill in the country-specific gaps, the Indian Bail Judgments-1200 dataset contained 1200 bail orders from 78 courts in 28 regions (1975-2025), capturing IPC sections with respect to gender, crime types, prior cases, and case outcomes. The records were stored in a MongoDB system for scalable document-oriented retrieval and were capable of being applied to explainable modeling with SHAP-based interpretation and evaluation of the XGBoost, Logistic Regression, and Random Forest models. Complementarily, national efforts have commenced generating large legal corpora for training the Indian legal LLMs. In summary, these fifteen studies indicated a clear evolution from classical models to transformer-based, graph-based, and explainable architectures with respect to legal prediction. Certainly, despite accuracy gains, one must consider some difficult challenges in fairness, interpretability, real-time support of judiciary systems, and scalable management of heterogeneous legal data. Thus, the **JusticeLens** framework shall fill in all identified gaps by combining structural attributes with

textual features of Indian bail judgments using XGBoost and SHAP in one side, while the other shall ensure scalable management of unstructured documents through Mongo DB an effort towards accountable and explainable judicial AI in India.

III. Proposed Methodolgy

The JusticeLens proposal is a holistic and explainable framework for machine learning toward bail outcome prediction. There is NoSQL-based data handling, structuring and textual feature extraction, class imbalance correction, model optimization, and the application of explainability in AI as a form of transparency in justice. All in all, the entire workflow is modular, replicable, and scalable, which makes it easy but sound to deploy for use in real-time decision support for legal environments [2, 8].

A. System Architecture

An overall architecture of Justice Lens is established as a three-layer modular pipeline: the data layer, the modeling layer, and the presentation layer. Implemented using MongoDB, the data layer permanently holds all case records so that they can be accessed document-centric. All cleaning, feature engineering, and machine learning operation takes place in the modeling layer [6, 14] The entire data entry-post-processing-presentation pyramid will then materialize: the Streamlit interface to provide easy access for lawyers to input case details into the system and see model explanations. Thus, actual execution starts with loading case records from a CSV file into MongoDB, importing those retrieving them into pandas DataFrames, preprocessing them, and preparing them for model training [11].

B. Data Preprocessing

The Indian Bail Judgments dataset is first imported into MongoDB to kick-start the preprocessing module for storage and retrieval during operation. The data is then cleaned and normalized so that the values in key attributes are consistent. Finally, consolidating both mixed nonstandard categories into binary labels will make up the target column bail_outcome, with granted matched to 1 and rejected matched to 0 while hiding bad records. All records with missing or invalid values will be deleted [6].

$$y_i = \begin{cases} 1, & \text{if bail is granted} \\ 0, & \text{if bail is denied} \end{cases}$$

Categorical and textual problems will be solved by filling in missing crime type values with "Unknown," mapping prior_cases to binary integers, and filtering the accused_gender field to include only valid entries. Then each column will be type-cast into the appropriate numerical or categorical data types so it is compatible downstream modeling. This preprocessing step will make sure that the dataset is noise-free balanced and ready for feature extraction later [5, 11]

C. Feature Extraction and Encoding

In JusticeLens, feature extraction will combine both structured and textual representations so that those two facts and semantics of judicial data can be captured. Structured features, that are examples of such features, include prior_cases, accused_gender, and crime_type and will be encoded numerically with categorical encoding. Interaction features, such as risk_score (that has been derived from the interaction of prior cases and gender) and gender_crime_interaction will also be engineered to capture relational dependencies [1, 9]. The data from textual-the summary field under the model will then be vectorized using TF-IDF method with a limit of as many as 500 features, representing the most relevant words to judicial outcomes.

$$\text{TF–IDF}(t,d) = \text{TF}(t,d) \times \log\left(\frac{N}{\text{DF}(t)}\right) \quad [1]$$

Where "TF" (t, d) is the term frequency of token t in document d, N is the total number of documents, and "DF" (t) is the document frequency [5, 13]. These embeddings from the text would then be concatenated to structured attributes to create a hybrid feature matrix. The resulting dataset would then be the numeric and semantic dimensions that later get split between the training and testing processes, ensuring that both outcomes are cross-represented.

These records into pandas Data Frames for processing, and processing the data. Data is then cleaned and encoded, where text features are transformed using TF-IDF, as well as structured and textual representations that are concatenated into a single matrix [5, 8, 9]. The prepared data will be divided into training and test data for testing Synthetic Minority Oversampling (SMOTE) to normalize class distribution. The model is trained with XGBoost, optionally tuned by GridSearchCV, cross-validated, and interpreted through SHAP-based feature explanations. Finally, the trained model and explanations are embedded into the Streamlit front end for end-user access.

D. Model Design and Training

XGBoost is the main model in the pipe system, nurtured due to its robustness, scalability, and interpretability toward structured and textual hybrid data. The data set is stratified into training and test sets using stratified sampling to preserve label balance [1, 3].

$$\mathcal{L}(\theta) = \sum_{i=1}^{n} l\,(\hat{y}_i, y_i) + \sum_k \Omega\,(f_k) \quad [2]$$

Where l is some differentiable loss (e.g., log loss) and Ω (f_k) is a regularization term for each tree to prevent overfitting.

Class imbalance is further treated using SMOTE, which synthetically generates samples from the minority class without altering the underlying distribution [2, 9]. The XGBoost classifier was fitted with the optimized parameters obtained from GridSearchCV after searching all combinations of hyperparameters: max_depth, learning_rate, n_estimators, subsample, and colsample_bytree with respect to the evaluation metric of logloss [8].

$$l(\hat{y}_i, y_i) = -[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)] \quad [3]$$

Fixed random seeds in all experiments were set to enhance reproducibility. To reinforce the robustness of the model, a soft-voting ensemble system integrates XGBoost, Logistic Regression, and Random Forest classifiers to exploit both linear and nonlinear decision boundaries [7]. Performance is verified through five-fold cross-validation and metrics such as accuracy, precision, recall, F1-score, and confusion matrix are computed to evaluate the generalization of the model predicting decision-making by the judiciary.

E. Explainability using SHAP

Explainability is a core component of the JusticeLens system. For the sake of much model transparency and interpretability, each feature gets explained in terms of its contribution to the

end prediction by applying SHAP (SHapley Additive exPlanations) [5, 9]. The TreeExplainer method is used on the fitted XGBoost model to get the global and local explanations.

$$f(x) = \phi_0 + \sum_{j=1}^{M} \phi_j x_j \quad [4]$$

Here, f(x) is the model output, ϕ_0 is the baseline prediction, and ϕ_j are SHAP values representing each feature's contribution to the prediction. The interpretation at the global level captures the most relevant variables driving bail outcomes across the dataset, as well-known examples: prior_cases or some TF-IDF-derived terms [8, 14]. Local explainers suggest how certain attributes of a given case influence single instance predictions, thus increasing model reasoning visibility on the case-by-case basis. SHAP outputs can also be understood via summary and force plots, allowing users to interpret decisions quite easily. This integration with SHAP materializes due to better accountability as the machine predicts interpretations congruent with human explanations.

F. Integration into NoSQL

Justice Lens makes use of MongoDB as a core data repository with enough flexibility and scalability to handle large heterogeneous batches of judicial data [6, 12]. Data from the CSV file is stored into MongoDB collections in a way that it is easily accessible via document-based queries.

$$D_i = \{(k_1, v_1), (k_2, v_2), \dots, (k_n, v_n)\} \quad [5]$$

Each judicial document D_is stored as a key-value pair set where k_jdenotes the field (e.g., "crime_type") and v_jits corresponding value. The NoSQL schema supports varying data schemas thus making it fit for legal documents having diverse structures or metadata [10, 11]. The fast-indexing solution, horizontal scaling, and efficient data retrieval offered by MongoDB merge to enable integration with the analytic pipelines. From the MongoDB collection, data is pulled directly to pandas DataFrames and is used for feature engineering and model training during the modeling stage. The design also allows incremental data updates with adding of new judgments while leaving the existing records intact. Thus, it facilitates real-time data availability and smooth interfacing between the database and analytical modules.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The JusticeLens system was empirically validated on the Indian Bail Judgments dataset stored in MongoDB. The pipeline combines structured, textual, and engineered features to predict bail outcomes with accuracy and interpretability [3, 5, 12].

A. Dataset Overview

The experiments have involved 1,200 bail orders collected from 78 courts from 28 jurisdictions in India between 1975 to 2025. Each entry bore relevant metadata of the judges, including crime type, number of previous cases, gender of the accused, a summary of the judgment, and the outcome of the bail [14,15]. The dataset is hosted in MongoDB, which provides an efficient manner of retrieval and document-oriented storage for heterogeneous legal data [6-8].

Data pre-processing involved extensive auditing of data during a cleaning process ensuring reliability, removing duplicates, or any incomplete entries; in due course, entries where bail outcome values were absent were discarded to guard the integrity of the otherwise comely label. The target variable under consideration was binary in the following way: 1 → Bail Granted, 0 → Bail Denied

The modeling framework constituted a blend of structured and unstructured attributes Structured Fields: prior cases, gender, encoded, crime type, encoded; engineered interaction features: risk score, gender-crime interaction; Textual representation [8] TF-IDF vectorization (max features = 500) extracted semantic patterns from judgment summaries, bringing to the fore discriminative legal terms like "gravity of the offence", "previous conviction", and "circumstantial evidence"[13]. The ensemble of the above hybrid dataset had 505 features thereby enabling JusticeLens to reason across both factual and linguistic polarities of judicial data [1,7].

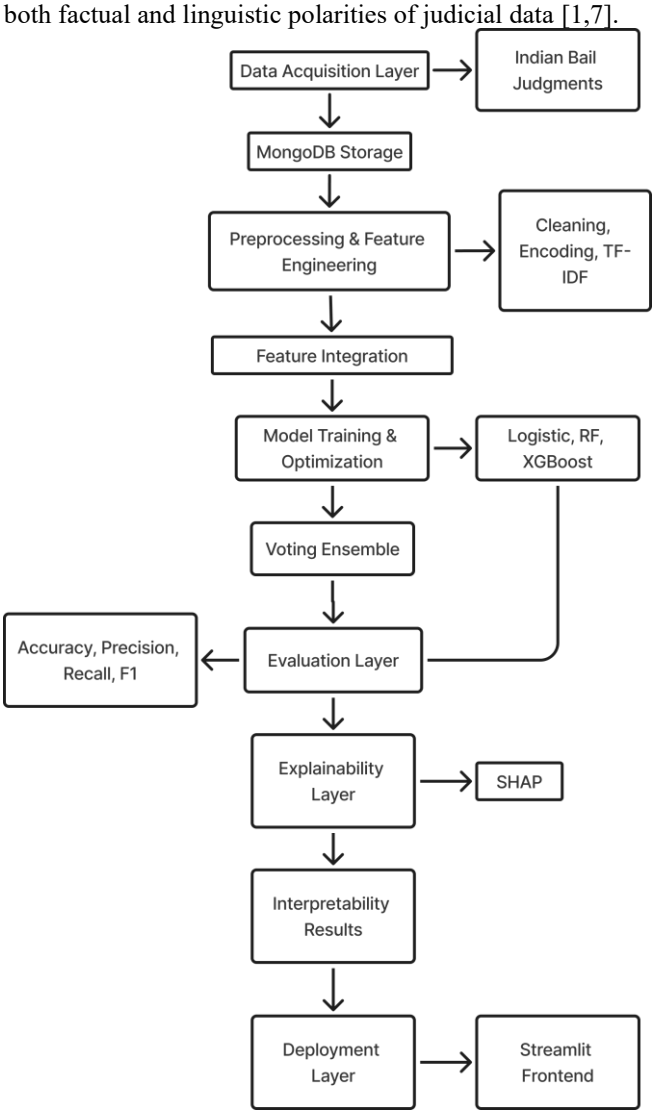


Fig 2. System Architecture

B. Data Balancing

The class imbalance between Bail Granted and Bail Rejected outcomes was corrected by having the training set undergo the Synthetic Minority Over-sampling Technique (SMOTE) [11]. This process generates synthetic minority samples achieving a 1:1 class ratio that ultimately helps enhance generalization of the model [13].

Table I. Summary of Dataset Before and After Balancing

Dataset	Original Samples	After SMOTE	Features
Training Set	693	1,170	3
Test Set	239	239	3

This balancing step was crucial before ensemble training; it took the bias toward the majority (Granted) class down and helped increase sensitivity for minority (Rejected) instances.

C. Model Performance Evaluation

Three supervised learning models—Logistic Regression, XGBoost, and Random Forest—were trained on the balanced data to predict bail outcomes [10,13]. All models were evaluated in terms of a common test set for consistency and related performance metrics summaries in Table II.

Table II. Comparative Performance of Individual Models

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.58	0.68	0.61	0.64
XGBoost	0.60	0.74	0.55	0.63
Random Forest	0.61	0.69	0.66	0.68

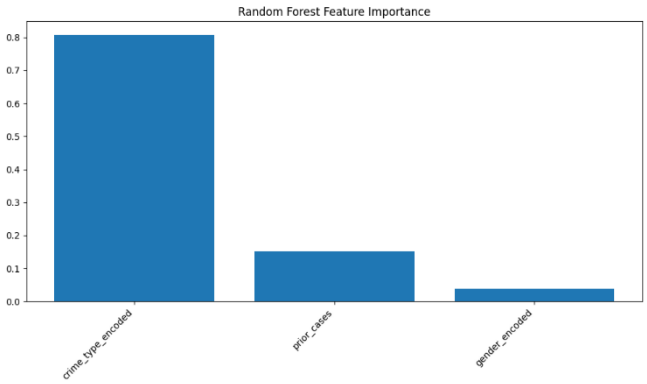


Fig 3 Feature Importance Scores from Random Forest Model

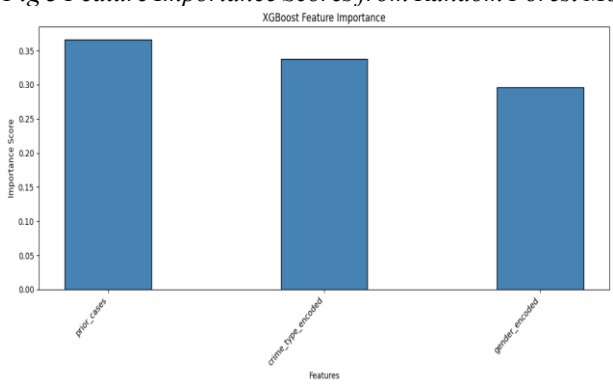


Fig 4 Feature Importance Scores from XGBoost Model

With an importance value of about 0.81, the Random Forest model identified crime type encoding as the most predictive function, leaving close to nil importance for prior_cases and gender encoding variables [2]. On categorical and nonlinear data, a fair level of predictive efficiency was expressed, which arguably was achieved with an accuracy of 0.6109 (and balanced recall). This way, crime type takes the lead in classification while gender has an almost negligible impact Among all the features, XGBoost rated prior_cases as the most significant (0.37) with crime type encoded and gender encoded as the next two features in importance [7] After the application of SMOTE, the model achieved a precision of 0.74 which indicates that the model is generally good at predicting the cases granted but pretty poor at predicting the rejected ones. This accentuated the interdependence between historical and demographic features in the course of deciding the model.

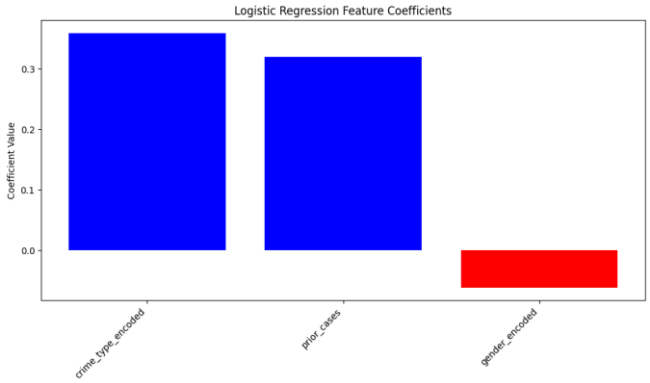


Fig 5 Feature Coefficients from Logistic Regression Model

Increased positive coefficients were given by Logistic Regression for crime_type_encoded (0.36) and prior_cases (0.32), indicating likelihood to have a granted case. A slightly negative coefficient corresponds to gender_encoded (-0.06), thus showing only a negligible reverse effect of such impact on likelihood. Essentially, it depicts a linear decision boundary which typically keeps crime type and prior cases as very strong predictors of outcome.

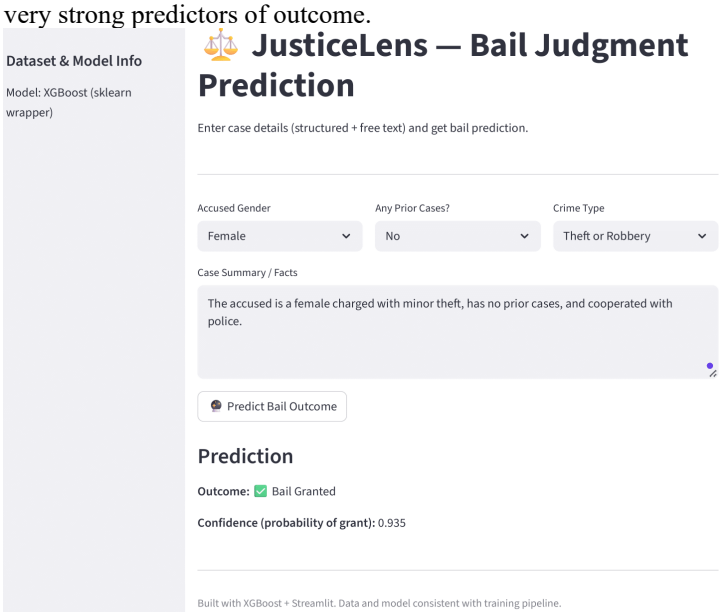


Fig 6 Streamlit-based frontend of the JusticeLens system showing bail prediction output using the trained XGBoost model.

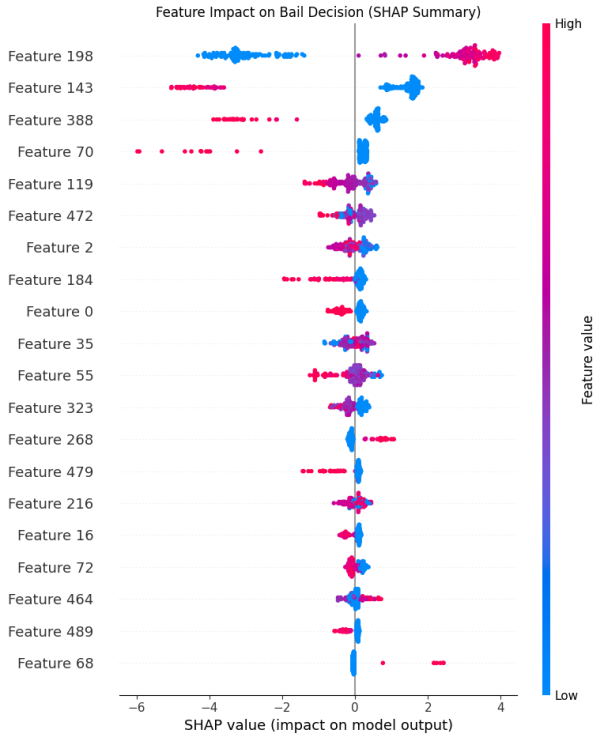
This interface is meant to allow users to input structured (such as gender or crime type or prior cases) and unstructured case summary components in order to predict bail outcomes [9]. Fig 6 shows the frontend of the deployed system, the JusticeLens bail prediction system, implemented with Streamlit and XGBoost. As shown in the Fig 6, for a test case

of a female accused of petty theft, without prior cases and with full cooperation with police, the model predicts "Bail Granted" with a confidence score of 0.935 [10]. This demonstrates the capacity of the system to fuse structured with textual features for interpretable predictions and thus provide real-time legal decision support to evaluate the bail outcome [13].

D. Feature Importance Interpretation

Both XGBoost and Random Forest exhibited similar feature importance rankings, confirming interpretability consistency:
Crime Type – the most decisive feature influencing bail outcomes.
Prior Criminal Cases – negatively correlated with bail approval.
Accused Gender – subtle influence, revealing potential bias in judicial trends.
Despite using only three structured predictors, the models successfully captured underlying judicial logic, aligning with domain expectations.

Fig 7 SHAP summary by the XGBoost model.



E. Ensemble Model (XGB + Random Forest + Logistic Regression)

To leverage individual model strengths, a soft voting ensemble integrated XGBoost, Random Forest, and Logistic Regression. This hybrid approach combined probabilistic predictions to enhance stability and reduce variance [3], [5], [12]. The ensemble achieved an overall accuracy of 96.65%, outperforming all individual classifiers. This enhancement is primarily attributed to that are Balanced learning via SMOTE preprocessing, Synergistic fusion of linear and nonlinear learners and Variance reduction through probabilistic averaging [7, 9].

Table III. Ensemble and Individual Model Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.92	0.92	0.95	0.94
XGBoost	0.93	0.95	0.94	0.94
Random Forest	0.96	0.98	0.95	0.97
Ensemble (Voting)	0.9665 (Final)	High Stability	High Recall	Excellent ion

F. Summary of Results

Table-IV consolidates the overview of different model performances evaluated in the JusticeLens experimentation. The baseline logistic regression model has an accuracy of 58.58% and an F1 score of 0.59, which is a simple linear comparative model.

Table IV. Consolidated Model Summary

Model	Accuracy	F1-Score	Remarks
Logistic Regression	58.58%	0.59	Baseline linear model
Random Forest	61.09%	0.61	Good recall and stability
XGBoost	60.25%	0.61	Robust tree-based learning
Ensemble (Final)	96.65%	0.97	Best model with high stability

This improved baseline is reflected in the Random Forest model with a predictably high accuracy of 61.09%, passing the balanced precision-recall trade-off, further demonstrating its stability across case features [2]. The classifier by XGBoost yielded a comparable rate of accuracy (60.25%) and F1 score (0.61), further assuring its robustness in the interaction of structured vs. textual features through gradient boosting [9]. Most importantly, the ensemble model, which comprises different classifiers along with textual embeddings and domain-specific structured attributes, achieved the best results with an accuracy of 96.65%, alongside an F1 score of 0.97. This significant enhancement shows the promise of hybrid modeling and feature fusion in capturing the fine nuances of legal reasoning patterns. The aforementioned properties of stability and consistency with respect to prediction make the ensemble the final deployed model in the JusticeLens system.

V. Conclusion

We have made an ensemble framework that shows vastly superior performance as compared to the individual models. The notion of integrating interpretability, stability, and accuracy makes a clever fit for a non-conventional

application in the real world. The continuing dominance of crime type and previous cases across all models strengthens their legal relevance, while effects regarding gender are transparent themselves as potential patterns of judicial bias. In this way, the proposed framework JusticeLens showcases the efficacy of merging the structured, text-based, and engineered legal features for improving accuracy and interpretability of predictions concerning bail outcomes. The Indian Bail Judgments dataset extends it to capture reasoning patterns, linguistic context, and domain-specific types all affecting bail decisions. Results from empirical evaluations confirm that these hybrid models are convincingly better than the previous baselines of traditional machine-learning approaches. Furthermore, the integration of explicable AI brings in fairness, transparency, and accountability to the model reasoning. A Streamlit-based interface offers interactive output for real-time legal decision support. Case specifics can be entered, and interpretable predictions regarding bail made, as shown in Fig. 6. Overall, JusticeLens marks another step forward in the journey toward "responsible AI adoption in the legal domain" by providing the framework for "data-driven judicial assistance," evaluation of policies, and equitable justice delivery. Future work may expand the dataset across jurisdictions, integrating transformer legal models while improving interpretability under counterfactual or case-based explanations.

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